

SKILLS O'CLOCK · EMBL HEIDELBERG · 5 MAY 2026

Talk to Your Pipeline

Using AI Assistants for Customisable & Reproducible
Bioinformatics with Nextflow

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5 May 2026 · 14:00 CEST · Seminar Room 1-202 + Online

Session Overview

- 0–5 min The Reproducibility Problem in Bioinformatics
- 5–15 min How Nextflow Thinks
- 15–45 min Live Demo: Run the Pipeline + Adapt with Copilot
- 45–55 min When to Trust AI
- 55–60 min Wrap-up & Template Handoff

01

The Reproducibility Problem in Bioinformatics

The challenge of reproducibility in bioinformatics

"RNA-seq reads were trimmed using Trimmomatic, aligned to hg38 with STAR, and quantified using featureCounts. Differential expression analysis was performed with DESeq2 in R using custom scripts."

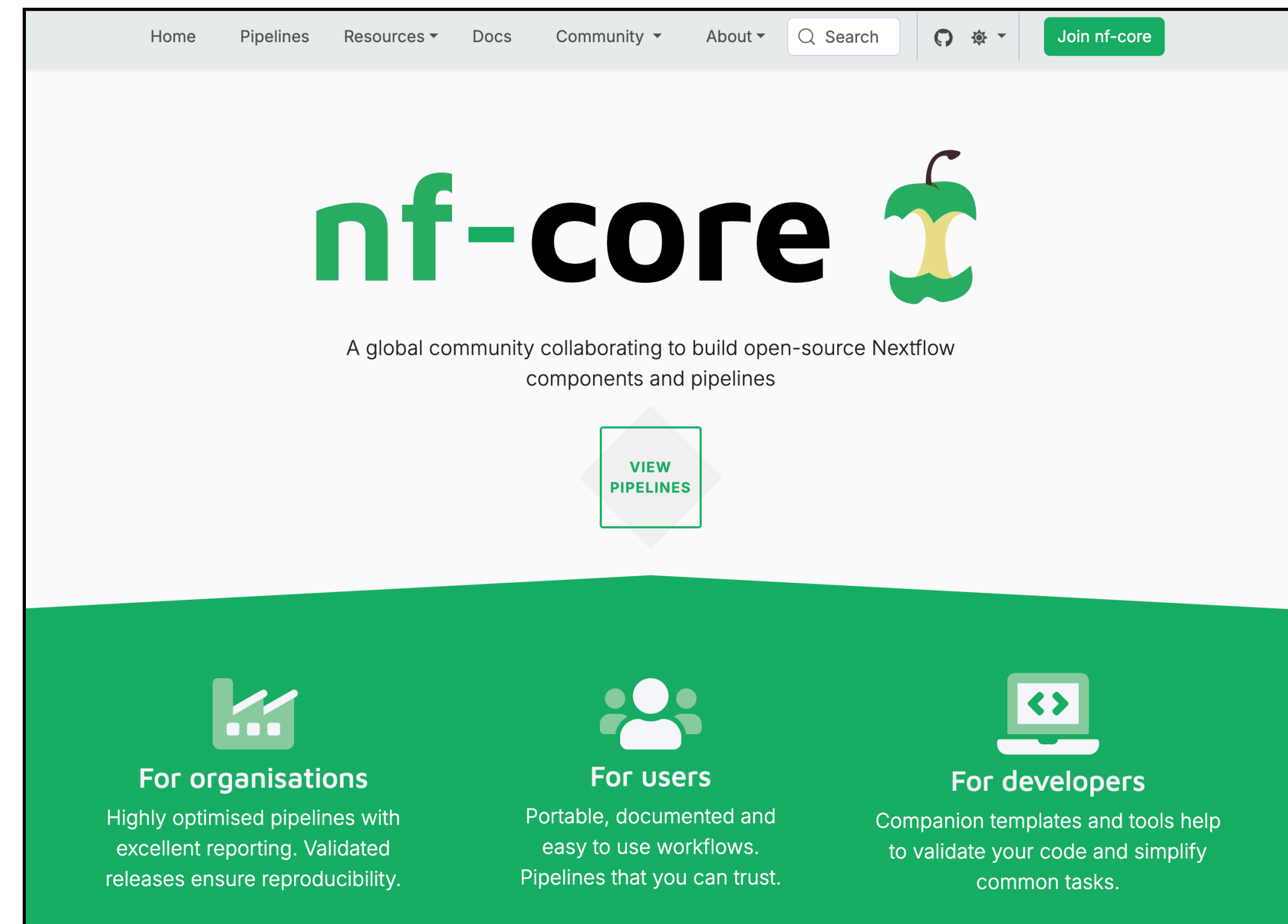
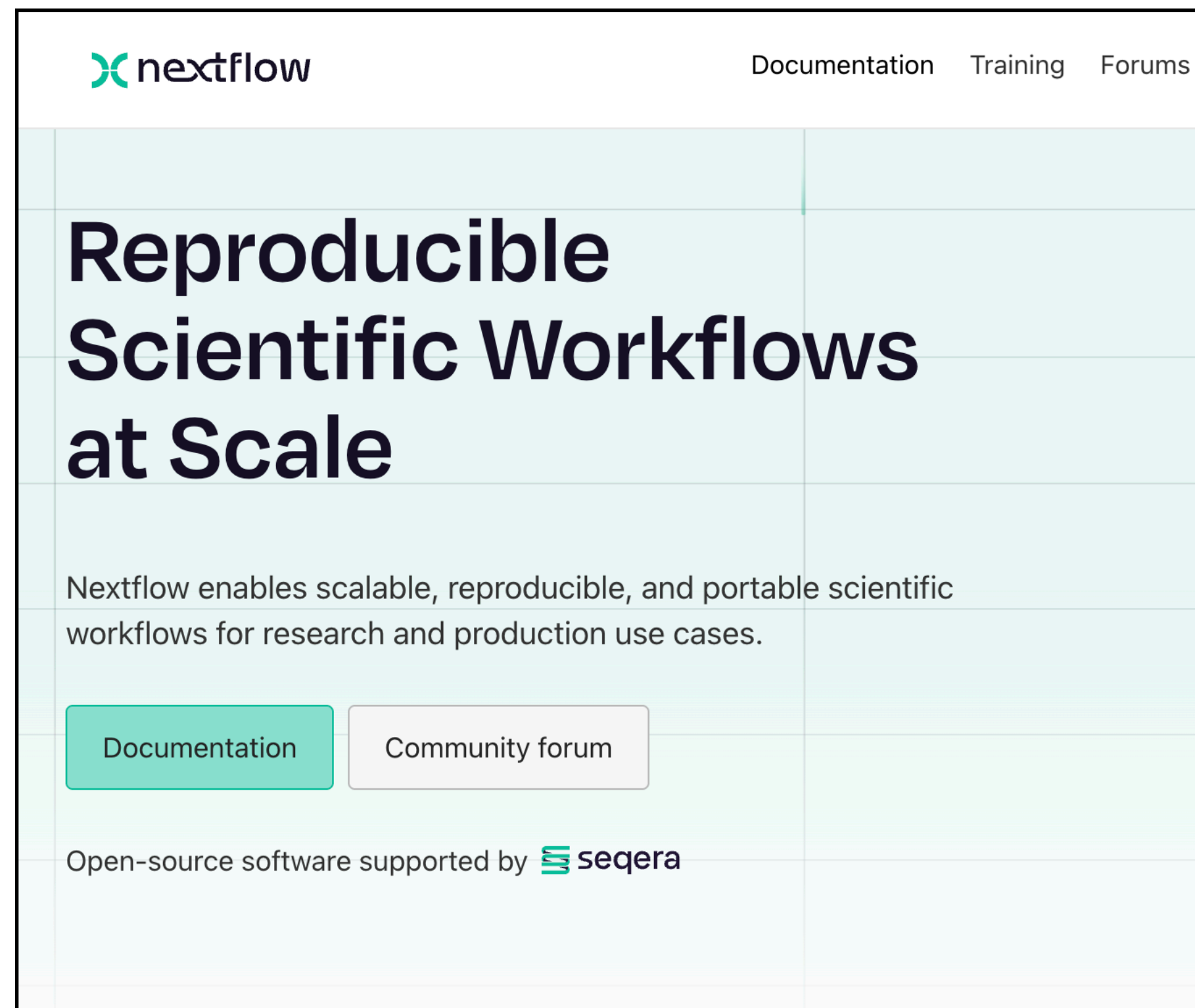
— typical Methods section of a paper

“Can you reproduce this?”

- Missing:
 - exact software versions
 - parameter values
 - often even whole scripts
- Reproducing these results would take 2 weeks of detective work (minimum)
- This is unfortunately the norm in bioinformatics

Nextflow + nf-core

- Pipeline-as-code: version-controlled, shareable, reproducible
- nf-core: community pipelines, rigorously tested across platforms
- Containers lock every software dependency to an exact version
- One command reproduces the exact same analysis on any machine



02

How Nextflow Thinks

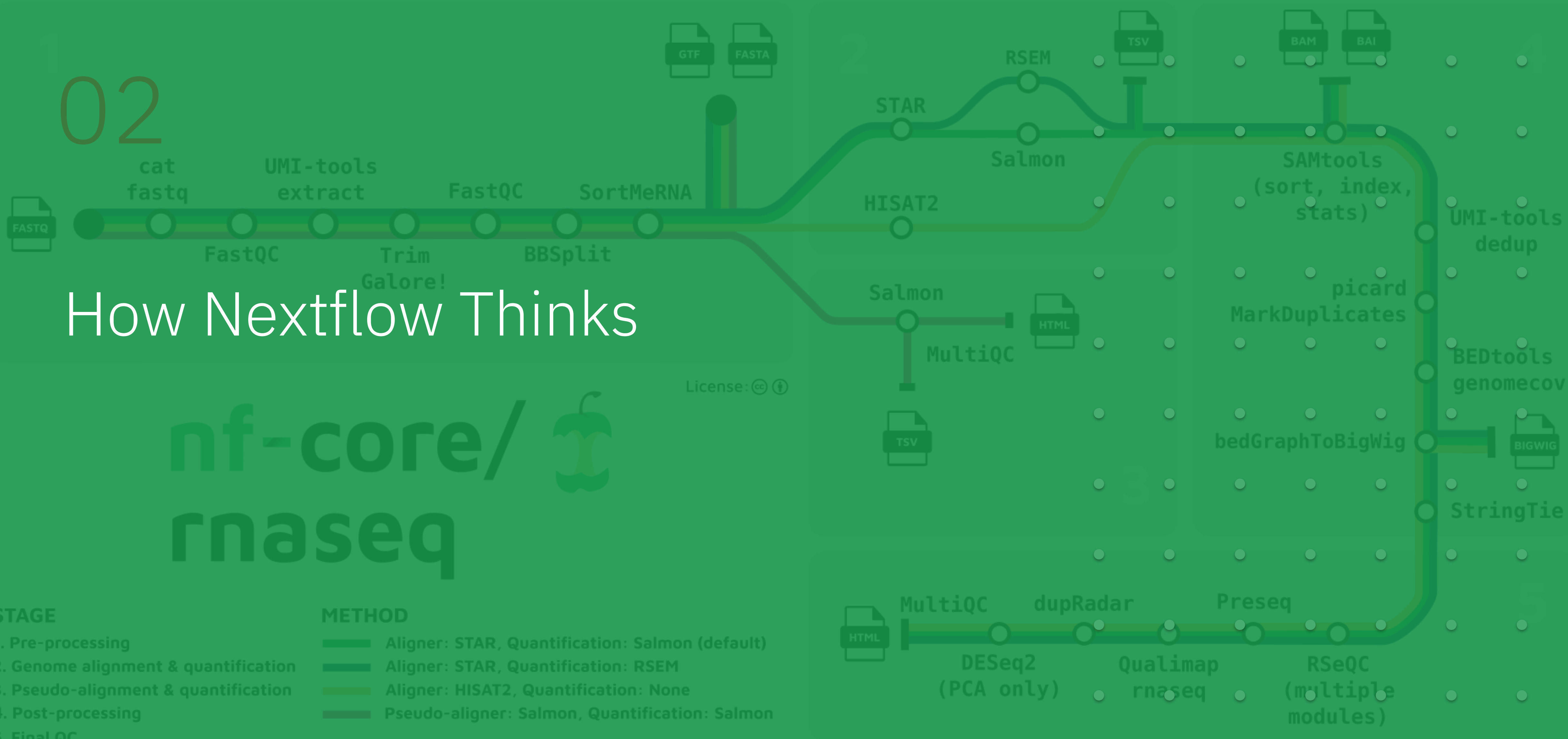
nf-core/
rnaseq

STAGE

1. Pre-processing
2. Genome alignment & quantification
3. Pseudo-alignment & quantification
4. Post-processing
5. Final QC

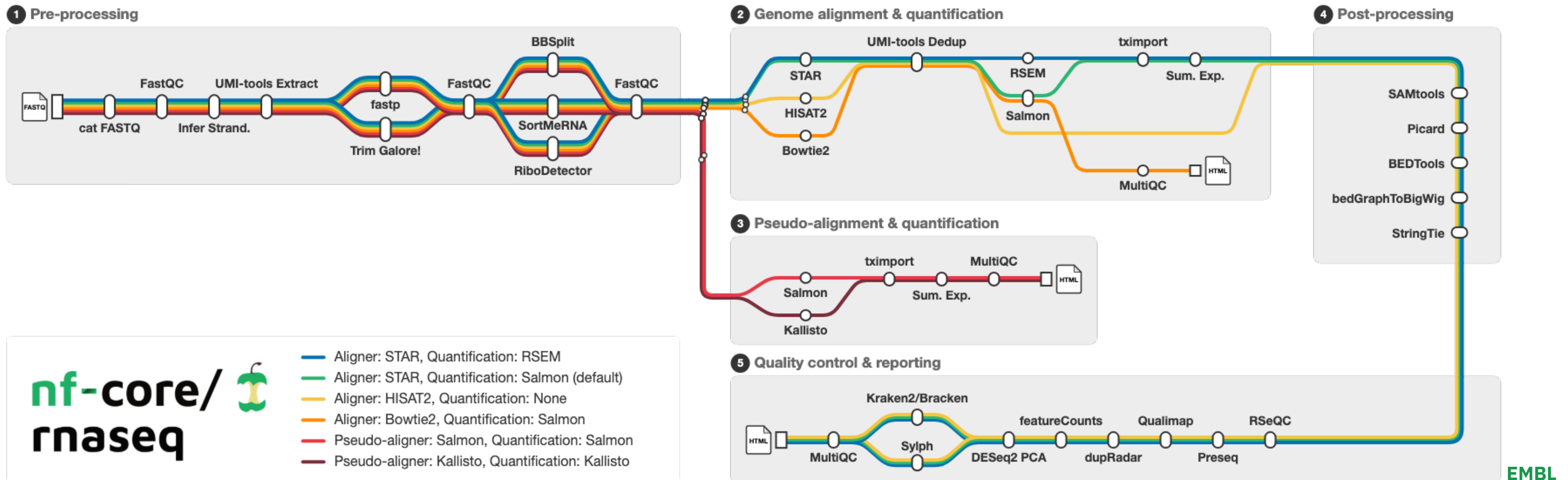
METHOD

- Aligner: STAR, Quantification: Salmon (default)
- Aligner: STAR, Quantification: RSEM
- Aligner: HISAT2, Quantification: None
- Pseudo-aligner: Salmon, Quantification: Salmon



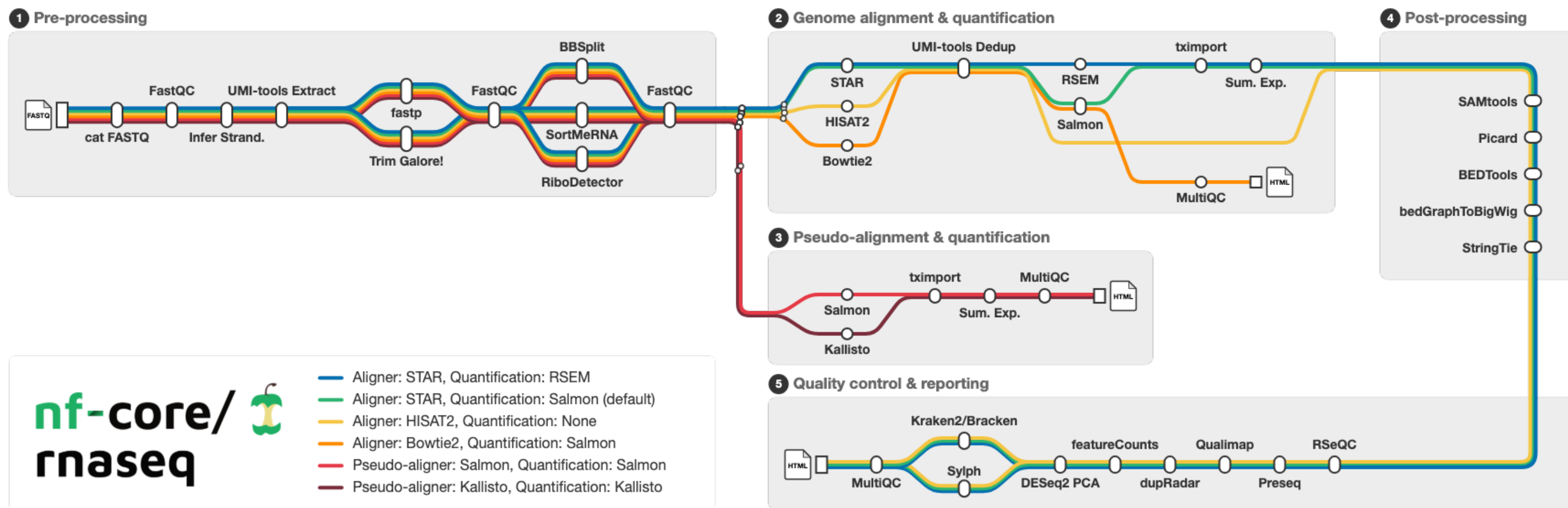
Individual steps — like functions

- Each process is self-contained: inputs in, command runs, outputs out
- Isolated by design — no hidden state, no side effects between steps
- Copilot can explain any process in plain English
- You don't need to write them, but you need to know how to read them



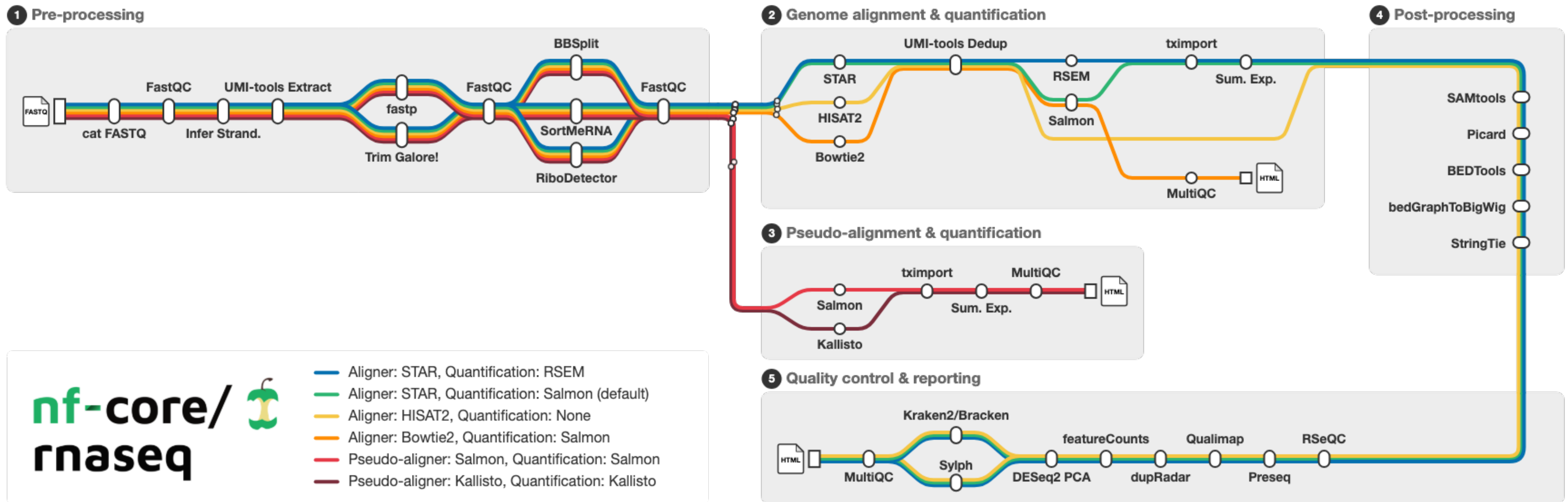
The pipes connecting processes

- Channels carry data between processes asynchronously
- Nextflow handles parallelism automatically through channels
- You rarely deal with channels directly - they are built to be scalable for larger datasets



The knobs you actually turn

- 90% of customisation lives here — no pipeline source code changes needed
- params block: genome, tools to skip, output directory, thread counts
- profiles block: docker, singularity, HPC cluster environments



03

The AI Landscape



GitHub Copilot

- Lives inside your editor — VS Code, JetBrains, Neovim
- Inline completions as you type + Copilot Chat for conversation
- Context: your current file and open tabs
- Best for: explaining code, writing boilerplate, fixing errors in the file you're looking at
- Access: \$10/month · free for students and open-source maintainers
- Made by: GitHub (Microsoft)

Cursor

- A VS Code fork with AI built in at a deeper level than an extension
- Composer mode: describe a change, it edits multiple files at once
- Context: the whole codebase, with semantic search over your repo
- Best for: refactoring across files, understanding unfamiliar codebases, multi-step edits
- Access: free tier available · \$20/month Pro · uses Claude, GPT-4o under the hood
- Made by: Anysphere

Claude Code

- Lives in your terminal — runs alongside your editor, not inside it
- Can read your entire codebase, run commands, and edit files autonomously
- Context: the whole project, not just the open file
- Best for: understanding large pipelines end-to-end, multi-file edits, asking 'how does this whole thing work?'
- Access: requires Claude subscription (claude.ai/code) or Anthropic API key
- Made by: Anthropic

OpenAI Codex

- OpenAI's agentic coding tool, launched April 2025
- Cloud-based: describe a task, it works on a sandboxed copy of your repo
- Can run tests, install packages, and open pull requests autonomously
- Best for: delegating longer tasks you don't want to supervise step by step
- Access: ChatGPT Plus / Pro (\$20–200/month)
- Made by: OpenAI
- Note: the original Codex model (2021) was retired in 2023 — this is a different product.

Choosing the right tool

- **Copilot:** in your editor, instant, great for line-by-line help
- **Cursor:** VS Code fork, codebase-aware, great for multi-file edits
- **Claude Code:** in your terminal, sees everything, great for whole-pipeline questions
- **Codex:** in the cloud, autonomous, great for delegating longer tasks
- These tools can be used in tandem with one another.
- Copilot is the most easily accessible - no new software required (works inside VS Code).

03

Run the Pipeline

Live Demo · 15–45 min



Two ways to run — quick, then real

- Part 1: nf-core test profile — 5–10 min, any laptop, no data download
- **Part 2: 4 real Himes et al (2014) samples, 200 K reads (on cluster)**
- Pre-run results: full 4-sample analysis ready to explore in MultiQC
- Same pipeline command for all three — only the params change

nf-core test profile: runs anywhere in 5 minutes

```
nextflow run nf-core/rnaseq \
  -profile test,docker \
  --outdir results/test
```

Pulls tiny FASTQ files from GitHub automatically — use this to verify the pipeline works before touching real data.

Himes et al. (2014): dexamethasone treated vs control

```
nextflow run nf-core/rnaseq \
  -profile slurm,singularity \
  -params-file params.yml \
  -c nextflow.config \
  -input samplesheet.csv \
  --genome GRCh38 \
  -outdir results
```

4 samples · 200 K reads each · GRCh38 ·

04

Deep-dive with Copilot

Live Demo · continues



What does this process do?

- Open `modules/nf-core/star/align/main.nf` in VS Code
- Ask in Copilot Chat: “Explain this Nextflow process in plain English”
- “What are its inputs and outputs?”
- (Low risk) You’re only asking for an explanation. Read the answer critically.

Write me a params.yml

- Open a blank my_params.yml in VS Code
- “Fill in this YAML file assuming I am running nf-core/rnaseq on mouse samples, GRCm39, skip dupradar”
- **Run:** nextflow run nf-core/rnaseq --params-file my_params.yml
- Always test on **-profile test** before applying to your real data

Paste the error, ask for the fix

- Copy the full error from the Nextflow terminal output
- Paste into Copilot Chat with context: pipeline, version, what you did
- Pattern: paste → understand → act (not googling blindly)
- Bonus: ask **Copilot** to generate a valid nf-core samplesheet CSV for a mouse dataset

05

When to Trust AI

Reflection · 45–55 min



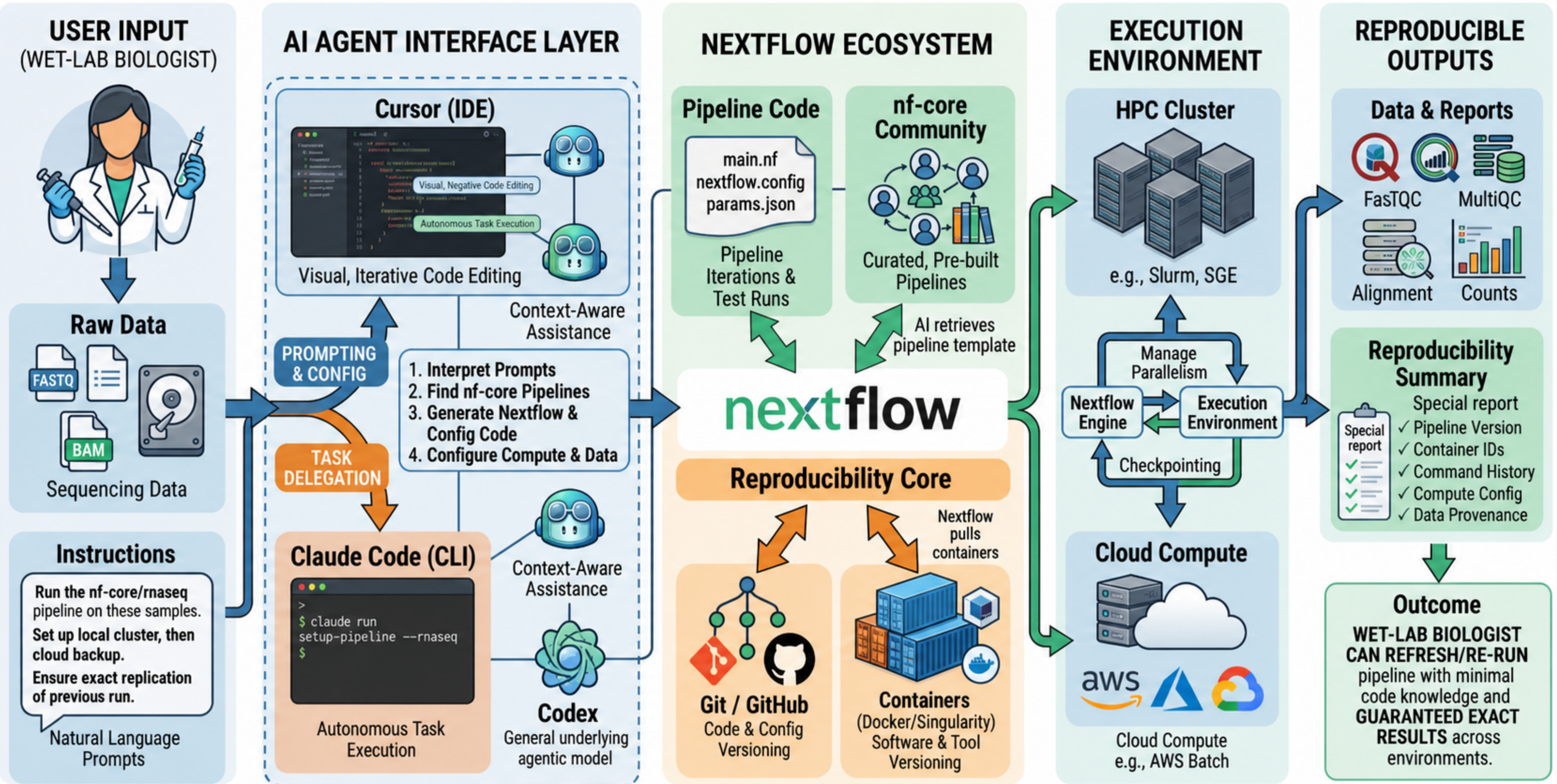
Know what AI is good at

- Trust it for explanation: asking what code does is very low risk
- Trust it for boilerplate: samplesheets, config blocks, params files
- **Don't trust it for biology**: it doesn't know your experimental design
- Always give the AI agent your context: assay, organism, pipeline version

Where AI confidently gets it wrong

- Alignment parameters without library type → wrong strandedness defaults
- Genome shorthand that doesn't exist in your nf-core version (version drift)
- **Important:** always provide full context to the AI agent before asking for fixes

AI-AGENT-ASSISTED WORKFLOW FOR WET-LAB BIOLOGISTS: REPRODUCIBLE NEXTFLOW PIPELINES



Questions?